

Zainelden Mahmoud

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EDUCATION

University of Toronto

Bachelor of Science, Computer Science Specialist, 4.0 GPA

Toronto, ON

September 2024 – June 2028

EXPERIENCE

Research Assistant

Far Data Lab

September 2026 - Present

Toronto, ON

- Incoming: investigating how fitting **Data Processing Units (DPUs)** with **KV caches** can improve their ability to perform **LLM inference**

Research Assistant

Toronto Intelligent Systems Lab

August 2026 - Present

Toronto, ON

- Incoming: investigating the effects of varying training dataset diversity on the reduction of shortcut learning in **Vision-Language-Action (VLA)** models

Software Developer

University of Toronto, Computer Science Student Union

August 2026 – Present

Toronto, ON

- Worked on terrain-diffusion, a recreation of the **InfiniteDiffusion** paper as a web app for interactive, diffusion-based terrain generation
- Built the model inference pipeline, orchestrating multi-stage diffusion generation into a coherent, servable pipeline
- Integrated coverage reports into the **CI/CD** pipeline using **coveralls.io**, leading to increased test case coverage

Machine Learning Engineer

eFinance Investment Group

June 2026 – September 2026

Cairo, EG

- Built an Egyptian national ID OCR pipeline to be used for eKYC authentication by the **Egyptian Stock Exchange**: post-trained **YOLO** model for field localization, **EasyOCR** for character recognition and fine-tuned **Qwen3-2B VLM** using **LoRA** as a confidence-gated fallback
- Amortized VLM inference cost by triggering it only below set confidence thresholds, keeping the primary pipeline CPU-only under limited GPU capacity

Software Developer

University of Toronto, Department of Computer Science

September 2025 – April 2026

Toronto, ON

- Worked on **PythonTA**, an open-source **Python** static analysis library used by **1500+** university students
- Implemented **Markdown** rendering for custom error messages and built renderers for multiple **pylint** error codes, improving code quality guidance across courses
- Discovered and patched an **XSS** vulnerability that allowed unsanitized user input to be injected into the **HTML** report renderer, potentially leading to remote code execution on university servers
- Fixed a memory leak caused by unbounded global cache growth, reducing average memory usage by **59%**

PROJECTS

CUDA Deep Learning Framework | C++, CUDA

- Built a **CUDA-accelerated** deep learning framework from scratch, implementing core neural network primitives
- Implemented **6 activation functions** (ReLU, Leaky ReLU, Tanh, GELU, Swish, Softmax) and their gradients
- Achieved up to **99.94% reduction** in runtime over CPU-based implementations through GPU parallelization
- Designed naive and tiled **shared-memory** matrix multiplication kernels, achieving up to **99.92% reduction** in runtime over a CPU baseline on 1024×1024 matmul

Distributed ML Parameter Server | C, Bash

- Built a distributed machine learning server to simulate a full training session for **logistic regression** using **SGD**
- Designed a non-blocking buffering system enabling the server to multiplex I/O across multiple concurrent clients
- Wrote **Bash** tests to simulate concurrent clients connecting to the server, increasing reliability and robustness

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, HTML/CSS, PostgreSQL, x64 assembly, Bash

Libraries & Frameworks: CUDA, Pandas, NumPy, scikit-learn, scapy, pwntools, FastAPI

Tools: Git, Unix/Linux, Binary Ninja, Wireshark, gdb